

JAVA PROGRAMMING LCA-2

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Course Name: Java Programming	Course Code:

1. Problem Statement:

Write a Java Program to implement Hierarchical Inheritance by creating one base class and two derived classes to display student and employee information.

2. Code:

```
// LCA-2 Java Programming: Sarthak Deodhar TY-CSEA (27, Batch B)
import java.util.Scanner;
class Person
{
    String name;
    int age;

    Person(String name, int age)
    {
        this.name = name;
        this.age = age;
    }

    void showPerson()
    {
        System.out.println("Name: " + name);
        System.out.println("Age: " + age);
    }
}

class Student extends Person
{
    int roll;
    String course;

    Student(String name, int age, int roll, String course)
    {
        super(name, age);
        this.roll = roll;
        this.course = course;
    }

    void showStudent()
    {
        System.out.println("Details Of Student:");
        showPerson();
        System.out.println("Roll No: " + roll);
    }
}
```

```

        System.out.println("Course: " + course);
    }
}

class Employee extends Person
{
    int empId;
    String dept;

    Employee(String name, int age, int empId, String dept)
    {
        super(name, age);
        this.empId = empId;
        this.dept = dept;
    }

    void showEmployee()
    {
        System.out.println("Details Of Employee");
        showPerson();
        System.out.println("Emp ID: " + empId);
        System.out.println("Dept: " + dept);
    }
}

public class hierarchical {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println("Enter Student name: ");
        String n = sc.nextLine();
        System.out.println("Enter Student Age: ");
        int a = sc.nextInt();
        System.out.println("Enter Student RollNo: ");
        int r = sc.nextInt();
        sc.nextLine();
        System.out.println("Enter Student Course: ");
        String c = sc.nextLine();

        Student s = new Student(n, a, r, c);

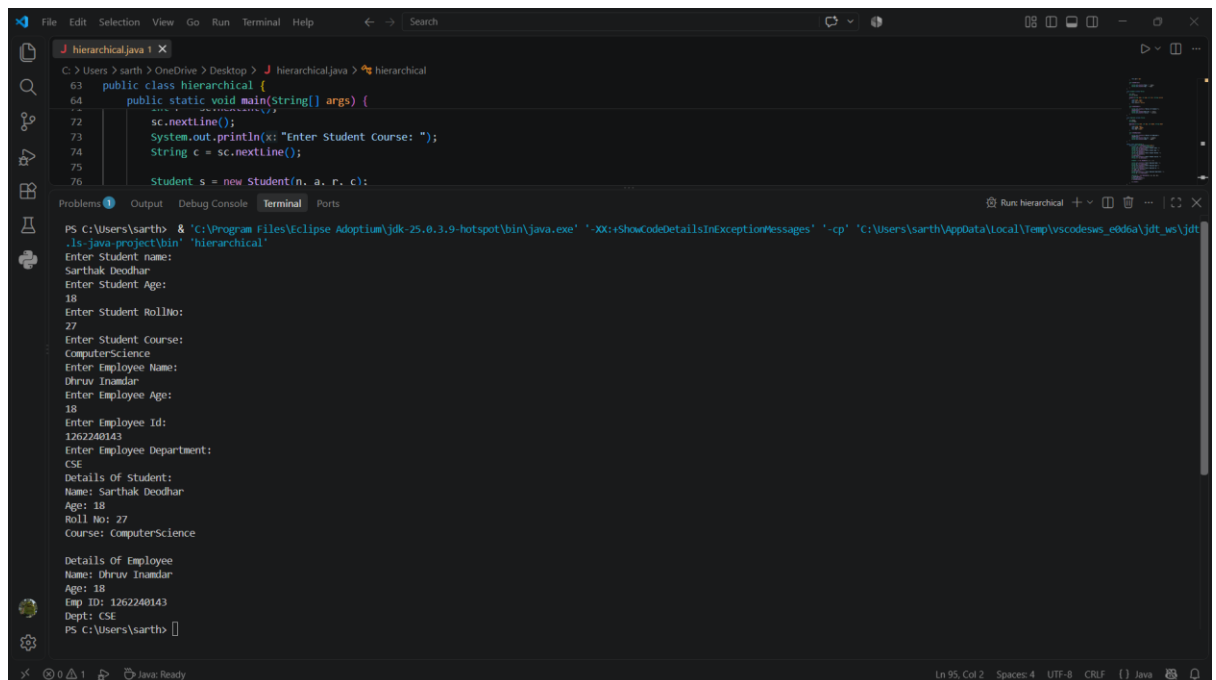
        System.out.println("Enter Employee Name: ");
        String n1 = sc.nextLine();
        System.out.println("Enter Employee Age:");
        int a1 = sc.nextInt();
        System.out.println("Enter Employee Id: ");
        int id1 = sc.nextInt();
        sc.nextLine();
        System.out.println("Enter Employee Department: ");
        String d1 = sc.nextLine();

        Employee e = new Employee(n1, a1, id1, d1);
        s.showStudent();
        System.out.println();
        e.showEmployee();

        sc.close();
    }
}

```

3. Output:



```
hierarchical.java
C:\Users\sarth> cd C:\OneDrive\Desktop> cd hierarchical
63 public class hierarchical {
64     public static void main(String[] args) {
65         Scanner sc = new Scanner(System.in);
66         System.out.println("Enter Student Name:");
67         String n = sc.nextLine();
68         System.out.println("Enter Student Age:");
69         int a = sc.nextInt();
70         System.out.println("Enter Student Roll No:");
71         int r = sc.nextInt();
72         System.out.println("Enter Student Course:");
73         String c = sc.nextLine();
74         Student s = new Student(n, a, r, c);
75     }
76 }
```

```
PS C:\Users\sarth> & 'C:\Program Files\Eclipse Adoptium\jdk-25.0.3.9-hotspot\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\sarth\AppData\Local\Temp\vscodesws_e0d6a\jdt_ws\jdt_ws\bin' 'C:\Users\sarth\AppData\Local\Temp\vscodesws_e0d6a\jdt_ws\jdt_ws\bin\hierarchical.jar'
Enter Student Name:
Sarathak Deodhar
Enter Student Age:
18
Enter Student Roll No:
27
Enter Student Course:
ComputerScience
Enter Employee Name:
Dhruv Inamdar
Enter Employee Age:
18
Enter Employee ID:
1262240143
Enter Employee Department:
CSE
Details Of Student:
Name: Sarathak Deodhar
Age: 18
Roll No: 27
Course: ComputerScience

Details Of Employee
Name: Dhruv Inamdar
Age: 18
Emp ID: 1262240143
Dept: CSE
PS C:\Users\sarth>
```

4. Code Logic:

Implementing Hierarchical Inheritance using 1 base class named Person, where basic details like name and age are taken. 2 Derived Classes Student and Employee both extends the class Person hence creating 1 Base class and 2 Derived class Hierarchical structure. In Student class, roll number and course are taken as input and then student details are displayed. In the second derived class Employee, Department and employee ID are taken as input and Displayed the Employee Details.

5. Github Link:

https://github.com/Sarthak23Deodhar/27_SarthakDeodhar_JavaProgramming_LCA-2